STAT 341: Tutorial Test 1 - Sample 1

First name ....

Last name ....

Signature ....

ID Number ....

Email ....

Time allowed: 50 minutes

Special instructions

- All numeric answers should be written either as a number to three decimal places or as a fraction reduced to its lowest terms.
- If working is not specifically requested, a correct answer will receive full marks. However, an incorrect answer could still receive credit if working is provided. An incorrect answer with no working will receive zero marks.

Please read: this practice test is designed to give you a clear sense of the sort of things that can come up in exams. Please note that this is, of course, only a subset of the material that could be covered. I would **strongly** encourage you to attempt this test under exam conditions - do not be tempted to look up something in your notes until afterwards! This is one reason why it is common for students to find real exams much harder than practice ones, even though the material is of an equivalent difficulty!

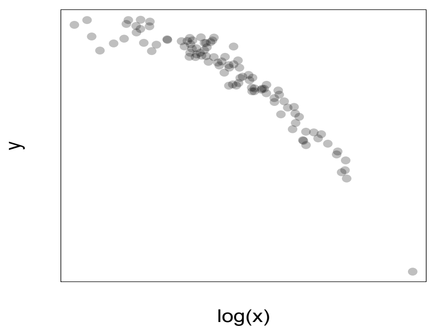
1. **(6 marks)** Location and scale equivariance.

   (a) *(3 marks)* Show that the population median is location-scale equivariant.

   (b) *(3 marks)* Show that the ratio of the population average to the population median is scale invariant but is neither location invariant nor location equivariant.

2. **(5 marks)** Power transformations.

(a) *(2 marks)* Consider the scatterplot below of y versus $\log x$:



Which of the following transformations might straighten this scatterplot?

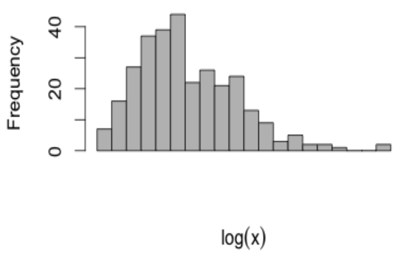
Circle those that **do apply**.

Cross out those that **do not**.

(i) $log(y)$ vs $log(x)$ (ii) y^2 vs $log(x)$

(iii) $log(y)$ vs x^2 (iv) y vs $\sqrt{x}$

(b) *(2 marks)* Consider the histogram below of $\log x$:



Which of the following transformations might make this histogram more symmetric?

Circle those that **do apply**.

Cross out those that **do not**.

(i) $\sqrt{x}$ (ii) $-\frac{1}{x}$

(iii) x^2 (iv) x

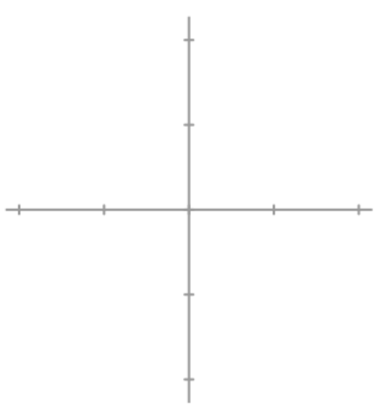
(c) *(1 mark)* Why do power transformations preserve the ranks of a population?

3. **(5 marks)** Consider the location statistic:

$$T_N(y_1, \ldots, y_N) = \frac{1}{2}\left(\min_i \ (y_i) \quad + \quad \max_i \ (y_i)\right) = \frac{y_{(1)} + y_{(N)}}{2}$$

(a) *(3 marks)* Derive this statistic's sensitivity curve:

(b) *(1 mark)* Give a sketch of the sensitivity curve:



(c) *(1 mark)* From this curve, what do you conclude about this statistic's resistance to outliers in y?

4. **(6 marks)** Implicitly defined attributes.

   (a) *(1 mark)* Give an example of an implicitly defined population attribute.

   (b) *(2 marks)* What is meant by a **location-scale equivariant** statistic?

   (c) *(3 marks)* In point form, describe the gradient descent algorithm.

5. **[5 marks]** Determine whether the following statements are true or false.

(a) The ratio of two scale equivariant attributes is a scale invariant attribute.

i. True

ii. False

(b) If $\alpha_1(\mathcal{P})$ is a location invariant attribute and $\alpha_2(\mathcal{P})$ is a location equivariant attribute, then $\alpha(\mathcal{P}) = \alpha_1(\mathcal{P}) + \alpha_2(\mathcal{P})$ is a location equivariant attribute.

i. True

ii. False

(c) Consider the population $\mathcal{P} = \{(x_1, y_1), ..., (x_N, y_N)\}$. If the histograms of transformed variates x^{α_1} and y^{α_2} are more or less symmetric, then the scatterplot of x^{α_1} versus y^{α_2} exhibits a more or less linear relationship.

i. True

ii. False

(d) It is preferred for a spread attribute to be location-scale invariant.

i. True

ii. False

(e) A skeweness attribute (a measure of asymmetry) should be both scale and location invariant.

i. True

ii. False